

Lecture 4: Broadband Media Delivery Standards, Formats, Protocols and Data Types

In this lecture, we continue the dissection of protocols, formats and data types used for media delivery. The focus now shifts from broadcast to broadband media delivery. Broadband delivery means TV and media delivery over IP networks.

The familiar three-stage model still applies: production, distribution and consumption. In previous lectures, we used this model to understand the overall media pipeline, the concrete production and consumption blocks, and the broadcast standards mapped to those blocks. Now we apply the same approach to broadband delivery.

Broadband distribution can be divided into two major types: managed and non-managed. Managed broadband is usually associated with IPTV services operated by telecom providers. Non-managed broadband is usually called Internet TV or Over-The-Top TV, abbreviated as OTT.

Managed and Non-Managed Broadband Delivery

Managed broadband means that the IP infrastructure is fully owned, leased or controlled by the distributor. This includes switches, gateways, wiring and other network equipment. A typical example is a telecom provider offering IPTV as part of a subscription package.

Non-managed broadband, or OTT, uses the public Internet to access content. The quality of delivery depends on the connection and infrastructure available at the viewer's current location.

In general, IPTV is more oriented toward fixed, living-room-style reception on TV receivers. It offers higher Quality of Service at the expense of flexibility. OTT is much more flexible: users can watch from many devices, on the go and from many locations. However, this flexibility comes with less predictable quality and availability. Buffering and resolution adaptation are common in OTT services such as YouTube or Netflix, especially on mobile or unstable networks.

Because of this, OTT depends heavily on adaptive streaming technologies, as introduced in Lecture 2.

Managed IPTV Protocol Stack

Managed IPTV is built on IP connectivity, reusing the well-established foundations of computer networking.

Each device in an IP network is uniquely identified by an IP address. Normally, one device sends data from its IP address to another device at a different IP address. This is one-to-one delivery.

However, IP also supports multicast delivery. In multicast, an IP address can represent a group of receivers rather than a single device. A sender transmits packets to the multicast group IP address, and the network distributes those packets only to receivers that joined the group.

IP Multicast and IGMP

IP multicast is especially suitable for broadcast-like delivery, such as live TV. Instead of sending one separate stream to every viewer, the network sends one stream through the shared infrastructure and copies packets only where necessary to reach multiple receivers.

Routers maintain multicast groups and resolve multicast IP addresses to actual receiver destinations. The Internet Group Management Protocol, or IGMP, is used to create and maintain these multicast groups.

This preserves bandwidth compared to one-to-one distribution and is therefore suitable for managed IPTV live TV services.

UDP and TCP in IPTV

Above the IP layer, the transport layer commonly uses protocols already known from general computer networking.

UDP, or User Datagram Protocol, enables delivery using ports. Clients can bind to a port and listen for incoming traffic. In IPTV multicast use cases, UDP delivery feels similar to broadcasting: whichever receiver joins the multicast group and listens to the correct port receives the packets intended for that stream.

However, UDP does not guarantee delivery. If packets are lost due to connectivity problems, they are not retransmitted. This makes UDP efficient but unreliable.

TCP, or Transmission Control Protocol, is used for one-to-one communication. A client connects to a server using an IP address and port. TCP provides reliability through acknowledgements and retransmissions, making it better suited for on-demand use cases such as Video on Demand.

In summary, UDP is commonly suitable for live multicast delivery, while TCP is more suitable for reliable client-server delivery.

RTP and RTSP

Additional protocols may be placed on top of the transport layer, often at the session or application level.

RTP, or Real-time Transport Protocol, uses UDP for delivery but adds media-specific functionality. It supports ordering, timing and synchronization. The receiver can detect packet loss, reorder packets and maintain a playback clock using packet timestamps.

RTSP, or Real Time Streaming Protocol, is commonly used as a control layer alongside RTP. It allows the receiver to control the stream using commands such as Play, Pause and Resume. It can be understood as a remote-control protocol for RTP-based streaming.

Together, RTP and RTSP make UDP-based delivery more suitable for media streaming, although they still rely on a managed or permissive network environment.

Network Management: SNMP and CWMP

Managed IPTV also relies on network monitoring and management.

SNMP, or Simple Network Management Protocol, allows operators to monitor devices and observe network health parameters.

CWMP, or CPE WAN Management Protocol, is also known as TR-069. It allows operators to manage customer premises equipment, update device software, configure services and monitor device state.

With SNMP and CWMP, telecom providers can fine-tune IPTV delivery. If the network becomes overloaded, operators can add infrastructure, increase bandwidth allocation, adjust user grouping or change routing strategies.

The cost of this control is that the consumption side must also be controlled. IP address allocation, allowed ports, protocol variants and processing rules are operator-specific. Users typically need either a dedicated set-top box or specialized software installed on a supported device.

This is why managed IPTV is usually not as simple as installing an app on any phone and watching from anywhere. The user often needs to be connected to the operator's specific network, usually at home.

Another limitation is that UDP and RTP traffic may be blocked by networks outside the operator's managed domain. This makes IPTV service maintenance difficult when multiple network providers are involved. Managed IPTV works best when the operator owns or controls the complete delivery chain.

Therefore, IPTV is commonly used in fixed environments: a dedicated router, a set-top box and a living-room TV.

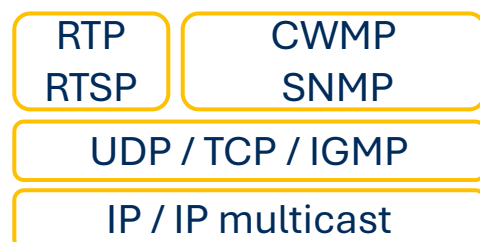


Figure 1. Managed IPTV protocol stack

Container Formats in Managed IPTV

Because managed IPTV is under operator control, it often reuses container formats from broadcasting.

MPEG Transport Stream is commonly used. In many installations, TS packets are simply repacked into RTP packets and sent across the IP network. The receiver unpacks RTP payloads and interprets the result as an MPEG transport stream.

From that point onward, processing can remain identical to broadcast TV. PES packets, signalling tables, audio and video coding formats and subtitles can all use the same structures discussed in the previous lecture.

This is one reason managed IPTV integrates well with broadcast receiver architectures.

Metadata in Managed IPTV

Metadata in managed IPTV may still be delivered inside the transport stream, as in broadcasting. Some operators use this approach.

However, the bidirectional nature of IP makes it attractive to deliver metadata separately. Instead of extracting all metadata from TS tables, the client can simply request metadata from a backend server.

This is commonly done over TCP/IP, often using HTTP and REST APIs. The client sends HTTP GET or POST requests and receives responses formatted as JSON or XML.

For example, when a device or app starts, it may download service lists, PIDs, service descriptions and other metadata from a backend server.

Video, Audio and Other Formats in Managed IPTV

Managed IPTV often uses the same video coding standards as broadcasting, especially when MPEG-TS is reused. AVC, HEVC and VVC are common examples.

This compatibility is useful because broadcast chips in set-top boxes and TVs are already designed to handle MPEG-TS, PES, common metadata structures and MPEG-family video codecs.

This makes IPTV a natural extension of a broadcast set-top box. Many operators use hybrid boxes that combine cable or satellite TV with managed IP services.

Audio formats are also similar to broadcast systems. AAC, AC-3 and surround formats are commonly used.

Other stream formats may also mirror broadcast TV, including DVB subtitles and teletext.

Content Protection in Managed IPTV

Because IPTV uses the IP protocol stack, operators can sometimes choose simpler deployment paths than those used in classical broadcasting.

Software-based Digital Rights Management, or DRM, frameworks are commonly used. Major examples include Google Widevine, Microsoft PlayReady and Apple FairPlay.

Devices may support multiple DRM schemes through different software libraries.

However, because managed IPTV networks are controlled, operators can also use vendor conditional access systems with custom set-top boxes, similar to broadcasting. Hybrid receivers may support both DRM and CAS, depending on the content and delivery mode.

SoCs for Managed IPTV

The range of SoC manufacturers for managed IP receivers is broader than in classical broadcast TV.

If the receiver only needs to support IPTV functionality, many tasks can be handled by processors known from the general computing world. However, PC processors are often too expensive or inefficient for set-top boxes.

Therefore, manufacturers often use embedded platforms from vendors such as Qualcomm, Broadcom and other DTV SoC providers. In MPEG-TS-based managed IPTV cases, broadcast-oriented SoCs work very well because the processing after network reception resembles broadcast TV.

Non-Managed IP: Internet TV and OTT

Non-managed IP delivery is commonly called Internet TV or OTT TV. Compared to managed IPTV, the most important differences appear in distribution, containers, metadata, codecs and content protection.

OTT Distribution Stack

In OTT, RTP and similar streaming-specific transport layers are usually not used. Instead, OTT relies on the most common TCP/IP communication model.

The typical flow is simple client-server communication over TCP, often with HTTP at the session or application level. Media data is delivered by downloading stream chunks.

At the application level, adaptive streaming protocols dominate. The two most important examples are Apple HLS, or HTTP Live Streaming, and MPEG-DASH, or Dynamic Adaptive Streaming over HTTP.

Both technologies use HTTP as their basis.

Why OTT Uses HTTP

OTT uses HTTP because of scalability and deployability.

Users access content through open networks and the public Internet. These networks are often restrictive. The traffic most reliably allowed everywhere is standard web traffic, especially HTTP and HTTPS over common ports.

Therefore, the easiest and most reliable approach is to reuse the web delivery model for media streaming. The client downloads media chunks in the same general way it downloads files from a web server.

This approach is not always optimal for bandwidth or live streaming latency, but it is extremely practical and scalable. Because network quality is unpredictable, adaptive streaming becomes essential.

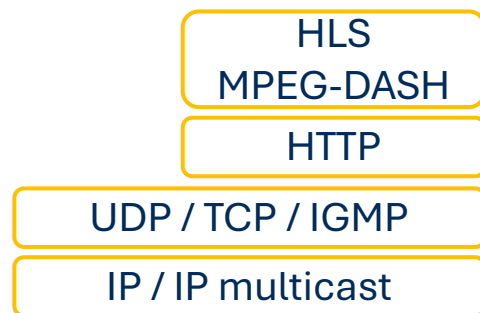


Figure 2. Internet TV / OTT protocol stack

OTT Container Formats

OTT extends the container world toward formats commonly supported by web browsers and web platforms.

One example is WebM, a container format based on Matroska.

The MPEG-driven approach is fragmented MP4, or fMP4. This is closely related to CMAF, the Common Media Application Format.

Unlike MPEG-TS packet-based delivery, OTT containers are based on file fragments within media tracks. These fragments can be individually downloaded as files over HTTP.

This makes them suitable for high-level, download-like delivery.

MPEG-TS may still be used in some OTT systems, but it must be placed into a file-based delivery structure. At the simplest level, a number of TS packets can be stored in a common file with no additional wrapping, and that file can be downloaded via HTTP.

OTT Metadata and Playlists

OTT metadata is typically delivered separately from media fragments.

Adaptive streaming requires playlist or manifest files. These are downloaded before playback begins and describe the stream structure.

Playlist files identify available chunks, quality levels, resolutions, bitrates and tracks. Based on this information, the player can formulate HTTP requests for the exact next chunk it needs, at the quality level suitable for current network conditions.

HLS uses M3U playlists.

MPEG-DASH uses MPD, or Media Presentation Description. MPD is more elaborate and describes the media presentation, periods, adaptation sets, representations and segment locations.

OTT Video and Audio Codecs

OTT systems can use the same codecs as managed IPTV and broadcast, such as AVC, HEVC and VVC. However, OTT also commonly uses web-oriented codecs.

VP9 is one important example. AV1 is its more standardized successor and is increasingly important in web-based streaming.

These codecs follow similar compression principles to MPEG-family codecs, but they are tuned for web distribution efficiency, quality and licensing considerations.

Audio formats are largely similar between managed and non-managed IP systems.

OTT Subtitles and Additional Data

OTT subtitles are commonly text-based.

Examples include WebVTT and TTML. These are separate text files that define subtitle text and timing. The receiver downloads them and renders the subtitles locally during playback.

This differs from DVB bitmap subtitles, where subtitle graphics are transmitted as bitmaps to be layered over video.

OTT Content Protection

In OTT, vendor-specific broadcast CAS is generally not used.

OTT must rely on mechanisms available across common consumer devices. Therefore, DRM systems dominate. Widevine, PlayReady and FairPlay are the common practical examples.

OTT content protection is closely tied to client software, operating system support, browser capabilities, secure media paths and license servers.

Managed IPTV Production Pipeline

We can now revisit the production pipeline for managed IPTV and map the formats to its phases.

A/V Coding

Audio and video coding is largely the same as in broadcasting. For example, MPEG AVC may be used, with groups of pictures, frames, macroblocks, motion vectors and residuals.

The coded data is packaged into PES packets and then into MPEG-TS in the subsequent phases.

Encryption

The encryption phase may differ from broadcasting because IPTV often relies on DRM or hybrid protection approaches.

For example, a custom encryption stack such as a vendor-specific system may receive the transport stream and output two protected streams: one for OTT or Internet TV CDNs using the selected DRM, and another for IPTV CDNs using custom scrambling.

Chunking in Managed IPTV

Classical chunking is usually not central in managed IPTV live delivery.

The transport stream may be stored as one or more files, but the file size mainly relates to stream availability or memory management. The stream is usually delivered by reading TS packets from memory and packing them into streaming packets.

For managed IPTV, there may be two provisioning server groups: one for video-on-demand content and another for live TV, often referred to as a CDN.

Managed IPTV Streaming Stack

In managed IPTV live delivery, the stream is buffered in memory in MPEG-TS format. TS packets are repacked as RTP packets. RTP packets are delivered as UDP datagrams through IP multicast.

IGMP allows users to subscribe to the multicast group intended to receive the selected content.

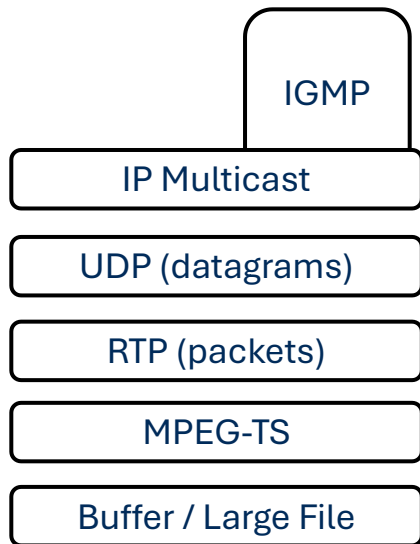


Figure 3. Managed IPTV streaming hierarchy

Backend and APIs

Managed IPTV also uses backend server groups such as OTT middleware provisioning and access management.

The backend may include remote monitoring features over CWMP or SNMP. Many service functions are exposed through REST APIs.

For example:

- A GET SERVICES method can return names and identifiers of available live TV services.
- A GET SERVICE INFO method can return the IP address and port used for streaming a specific service, along with technical information such as container format and PIDs.
- LOG IN and LOG OUT methods allow users to authenticate before accessing restricted services.

This backend combines media metadata, access control and operational management.

OTT Production Pipeline

The non-managed OTT production pipeline differs more significantly from broadcast and managed IPTV.

A/V Coding

OTT may use AVC, HEVC or VVC, but it also commonly uses codecs such as VP9 and AV1.

The conceptual coding principles are similar: frame groups, predicted pictures, residuals, motion vectors and compressed representations. The terminology and exact data structures differ by codec.

Program Stream Packaging and Fragmentation

In OTT, program stream packaging becomes increasingly different.

Each decodable unit, such as one or more frame groups, is packed as a fragment inside a dedicated media track. The fragmentation format may be based on CMAF.

Fragments across tracks may have different sizes, but they correspond to the same time slices and cover the same duration. For example, a video fragment and an audio fragment may represent the same two or four seconds of media.

These fragments are usually indexed by timestamps. Players use these timestamps to request the correct fragments as playback progresses.

The complete structure of chunks is stored on the server in a folder or path structure.

Classical multiplexing is therefore usually not used in OTT. Instead of one multiplexed stream containing all tracks and metadata, OTT commonly stores separate tracks and fragments that the client downloads as needed.

DRM and Common Encryption

OTT uses DRM on top of media fragments. The fragments are encrypted before delivery.

To make encryption interoperable across devices, Common Encryption, or CENC, is used. CENC defines a common way to encrypt media content, while DRM systems define how keys are formatted, delivered and licensed.

This allows the same encrypted media content to be used with different DRM systems, depending on the client device and platform.

Chunking and MPD

Chunking finalizes the fragmented stream structure.

In MPEG-DASH, the playlist or manifest is the MPD file. MPD describes the complete media presentation.

An MPD may define periods. Periods are useful for separating different content intervals. In live TV, periods may represent different shows. Within a show, periods can also be used to separate advertisement intervals.

This makes ad insertion practical. A personalized ad period can be inserted by referencing period specifications from another MPD or by modifying the presentation structure for the target user.

Each period is further defined by adaptation sets. Each adaptation set usually corresponds to one media track type, such as video or audio.

Inside each adaptation set, there are multiple representations. Representations describe quality variants, such as different resolutions or bitrates.

With this information, the player can choose which representation to use for a given track and timestamp based on current network conditions.

As playback continues, the player requests fragments corresponding to the required timestamps and selected quality representations. These fragments are exactly the ones created during packaging.

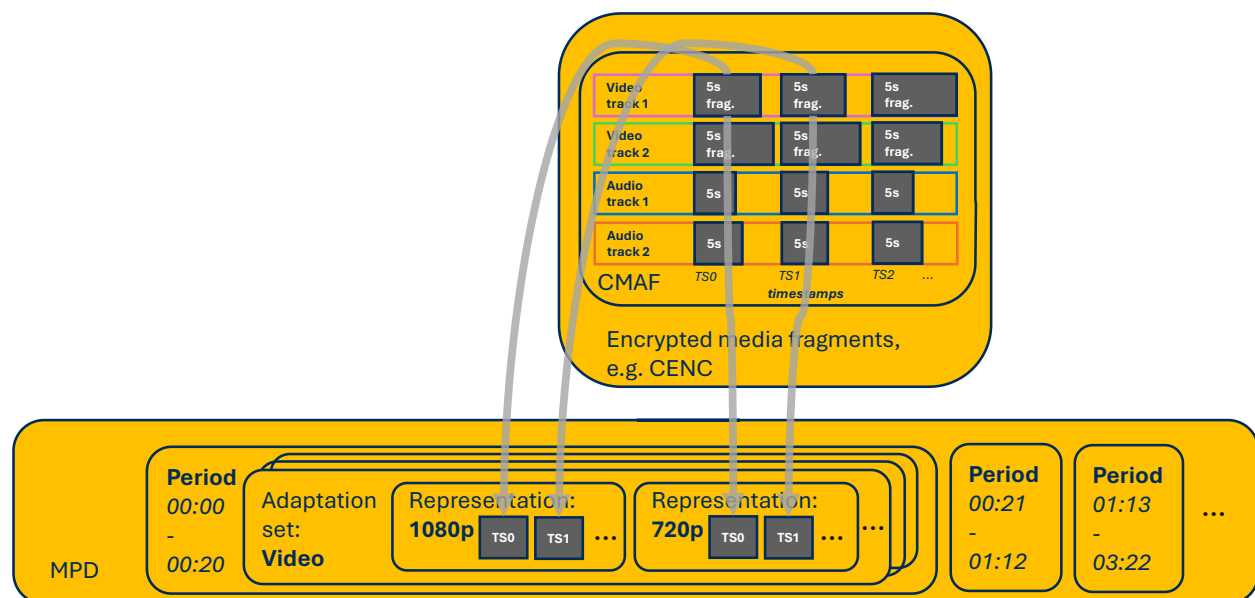


Figure 4. Chunking in OTT and referencing in the playlist

OTT Publishing

In OTT publishing, VoD and CDN servers serve playlist files and stream chunks.

This is less like continuous streaming and more like serving a structured set of files for download via HTTP.

HTTP always uses one-to-one delivery. The client requests a playlist or fragment, and the server responds with the requested data.

The transport may use TCP. It may also use UDP through QUIC, which was originally developed by Google as an optimization for HTTP-style delivery. Even when QUIC uses UDP underneath, the delivery model remains one-to-one HTTP-based communication, not broadcast or multicast.

All of this is based on the IP stack.

OTT Middleware and DRM Licensing APIs

OTT backend servers provide REST APIs for metadata delivery, service discovery and user management.

DRM licensing and key-related operations are also exposed through backend APIs. The client uses these APIs to obtain licenses or keys required to decrypt protected fragments.

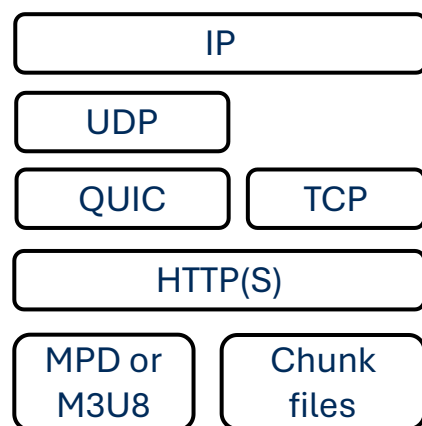


Figure 5. Internet TV / OTT streaming hierarchy

Recap: Broadband TV Standards and Formats

To summarize, broadband TV delivery relies on a well-defined stack of networking, streaming, container and media formats.

At the lowest transmission and management layer, the system is built on the TCP/IP stack. In managed IPTV environments, delivery often uses UDP over IP multicast for live TV or unicast for specific services, typically combined with protocols such as RTP. In contrast, OTT delivery relies on HTTP-based communication, using adaptive streaming technologies such as MPEG-DASH and HLS.

For device and network management in IPTV systems, protocols such as CWMP (TR-069) and SNMP are commonly used, enabling monitoring, configuration and maintenance of network devices and receivers.

In terms of container formats, managed IPTV frequently reuses broadcast formats such as MPEG-TS and PES. OTT systems, however, favor fragmented, file-based containers suited for HTTP delivery, such as fragmented MP4 (fMP4, often within CMAF) or WebM.

Metadata in broadband systems is predominantly delivered via HTTP-based REST APIs. Streaming-specific metadata is provided through playlist or manifest files—MPD in MPEG-DASH and M3U in HLS—while additional metadata is typically delivered in JSON or XML format within HTTP responses.

For media data itself, video formats largely follow those used in broadcasting, including AVC, HEVC and VVC, with the addition of web-oriented codecs such as VP9 and AV1. Audio formats remain consistent across delivery types, with AAC, HE-AAC, AC-3 and surround formats such as Dolby and DTS being widely used.

Subtitles in IPTV may still use DVB formats, but OTT systems more commonly rely on text-based formats such as WebVTT and TTML. Finally, application and interactive content in broadband environments align with general web standards, leveraging the full ecosystem of web technologies and applications that can be rendered on modern devices.

Data	Video: VP9, AV1, H.264 AVC, H.265 HEVC, H.266 VVC Audio: AAC, HE-AAC, AC3, surround (Dolby / DTS) Subtitles: WebVTT, TTML / IMSC, DVB-SU Apps: Internet content in general (W3C: HTML+CSS+JS ...)
Metadata	MPD M3U8 Custom JSON / XML over HTTP / REST
Containers	fMP4 / CMAF WebM / Matroska MPEG-TS / PES Protected via CENC + DRM keys and licensing scheme
Transmission and management	UDP over IP Multicast RTP over UDP over IP Multicast or Unicast HTTP Adaptive Streaming: MPEG-DASH, HLS TR-069 CWMP or SNMP for device and network management

Figure 6. Most common standards and formats in broadband TV

Applying Broadband Details to Project Work

With both broadcast and broadband delivery now covered, the project work can become more precise.

When revisiting a dataflow diagram, each line should be inspected and annotated with the relevant standard, format or data type.

For OTT cases such as Netflix, students should identify items such as:

- HTTP or HTTPS requests
- TCP/IP or HTTP over QUIC
- HLS playlists or MPEG-DASH MPD files
- CMAF or fMP4 fragments
- Audio and video tracks
- VP9, AV1, AVC, HEVC or other codec data
- WebVTT or TTML subtitle files
- DRM licenses and CENC-encrypted fragments
- JSON or XML metadata returned by REST APIs

The same applies to UML sequence diagrams. Commands and exchanges can now be linked to specific standards and parameters.

For example, the sequence diagram may show the app requesting service metadata from OTT middleware, downloading an MPD, selecting a representation based on bandwidth, requesting a fragment over HTTP, obtaining a DRM license, decrypting a CENC-protected fragment, and passing decoded audio and video toward presentation.

The goal is to make the architecture closer to an actual implementation, especially in adaptive streaming scenarios.

Closing Perspective

This lecture completed the standards-level view of broadband media delivery.

Managed IPTV uses IP networks under operator control and often reuses broadcast container structures such as MPEG-TS. Live IPTV commonly relies on IP multicast, UDP, RTP and IGMP, while service management may use REST APIs, SNMP and CWMP. Because the operator controls the network and often the receiver, quality can be high, but flexibility is limited.

OTT uses the public Internet and therefore relies on scalable, widely accepted web delivery mechanisms. HTTP-based adaptive streaming, through HLS or MPEG-DASH, dominates. Media is organized as downloadable chunks or fragments, with playlists or manifests describing the available tracks and quality levels. DRM systems protect content, commonly using CENC and license servers.

Together with the previous lecture on broadcast standards, this gives a complete foundation for describing Project 1 use cases in terms of real protocols, formats and data types.